Q1.

A particle, P, of mass M is released from rest and moves along a horizontal straight line through a point O. When P is at a displacement of x metres from O, moving with a speed v ms^{-1}, a force of magnitude $|8Mx|$ acts on the particle directed towards O. A resistive force, of magnitude $4Mv$, also acts on P.

(a) Initially P is held at rest at a displacement of 1 metre from O. Describe completely the motion of P after it is released.

Fully justify your answer.

(8)

(b) It is decided to alter the resistive force so that the motion of P is critically damped.

Determine the magnitude of the resistive force that will produce critically damped motion.

(4)

(Total 12 marks)

Q2.

An isolated island is populated by rabbits and foxes. At time t the number of rabbits is x and the number of foxes is y.

It is assumed that:

- The number of foxes increases at a rate proportional to the number of rabbits. When there are 200 rabbits the number of foxes is increasing at a rate of 20 foxes per unit period of time.
- If there were no foxes present, the number of rabbits would increase by 120% in a unit period of time.
- When both foxes and rabbits are present the foxes kill rabbits at a rate that is equal to 110% of the current number of foxes.
- At time t = 0, the number of foxes is 20 and the number of rabbits is 80.

(a) (i) Construct a mathematical model for the number of rabbits.

(9)

(ii) Use this model to show that the number of rabbits has doubled after approximately 0.7 units of time.

(1)

(b) Suggest one way in which the model that you have used for the number of rabbits could be refined.

(1)

(Total 11 marks)